Using Potential Variables in Studies of Genetic Effects

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April 24, 2016

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Introduction

One of the critical issues in genetic association studies is to evaluate association between genetic risk and human complex diseases.

$$logit\ P(Y = 1|G) = \alpha + \gamma X + \beta_1 G_1 + \cdots + \beta_m G_m$$

To test

$$H_0 : \beta_1 = \cdots = \beta_m = 0 \ \text{ vs } \ H_1 : \exists \beta_i \neq 0 (i = 1, \ldots, m)$$

Introduction

Traditional Methods: test statistics derived under a specified genetic model by assigning a particular set of scores to the genotypes (Score Test, Wald Test, Likelihood Ratio Test)

Robust Methods: test statistics derived under the unknown genetic model (Pearson's Chi-Squared Test, MERT, MAX, CLRT, MAX3, MIN2)

Penetrance and Genotype Relative Risks

Consider a diallelic marker M, typically a SNP with alleles A and a.

Let the three genotypes of M be $G_0 = aa$, $G_1 = aA$ and $G_2 = AA$. The allele frequencies in the population are denoted by $p = Pr(A)$ and $q = Pr(a) = 1 - p$. The population genotype probabilities of G_i are denoted by $g_i, i = 0, 1, 2$.

If Hardy-Weinberg proportions hold in the population,

$$g_0 = q^2, \ g_1 = 2pq, \ and \ g_2 = p^2$$

Penetrance and Genotype Relative Risks

The disease prevalence in the population is denoted by $k = Pr(case)$. The penetrance of a disease given a genotype is denoted by $f_i = Pr(case|G_i), i = 0, 1, 2$.

$$k = \sum_{i=0}^{2} Pr(G_i)Pr(case|G_i) = f_0 g_0 + f_1 g_1 + f_2 g_2.$$

Penetrance and Genotype Relative Risks

In case-control studies, cases and controls are retrospectively sampled from populations.

Table 1. Genotype distribution

	aa	aA	AA	$Total$
Cases	r_0	r_1	r_2	R
Controls	s_0	s_1	s_2	S
Total	n_0	n_1	n_2	N

Assume that $(r_0, r_1, r_2) \sim Multinomial(R; p_0, p_1, p_2)$ and that $(s_0, s_1, s_2) \sim Multinomial(S; q_0, q_1, q_2)$. $p_i = Pr(G_i|case)$ and $q_i = Pr(G_i|control)$ for $i = 0, 1, 2$.

Penetrance and Genotype Relative Risks

Denote Genotype Relative Risks(GRR) by $\lambda_i = f_i/f_0, i = 1, 2$, where $f_0 > 0$. Under the null hypothesis, the penetrances are equal to the prevalence,

$$f_0 = f_1 = f_2 = k \;\Leftrightarrow\; \lambda_1 = \lambda_2 = 1.$$

Genetic Models

A genetic model refers to a specific mode of inheritance.

- the recessive model(REC)
- the additive model(ADD)
- the multiplicative model(MUL)
- the dominant model(DOM)

Genetic Models

Assume that A is a high risk candidate allele and a is any of the other alleles. Then

$$f_2 \geq f_1 \geq f_0 \ and \ f_2 > f_0$$

- $f_1 = f_0$ or $\lambda_1 = 1$ (REC)
- $f_1 = (f_0 + f_2)/2$ or $\lambda_1 = (1 + \lambda_2)/2$ (ADD)
- $f_1 = \sqrt{f_0 f_2}$ or $\lambda_1 = (\lambda_2)^{1/2}$ (MUL)
- $f_1 = f_2$ or $\lambda_1 = \lambda_2$ (DOM)

The space Λ defines a family of genetic models indexed by $\theta \in [0, 1]$ as

$$\Lambda = \{(\lambda_1, \lambda_2) : \lambda_1 = (1 - \theta) + \theta\lambda_2, 0 \leq \theta \leq 1\}.$$

Our method

For many complex diseases, their underlying genetic models are often uncertain, that there is no single efficiency test for all scenarios.

$$logit\ P(Y = 1|G) = \alpha + \gamma X + \beta G$$

The genetic effect at a marker locus is coded by 0 for G_0, θ for G_1 and 1 for G_2, where $\theta \in [0, 1]$, and $G \in G_0, G_1, G_2$.

To test

$$H_0 : \beta = 0 \quad \text{vs} \quad H_1 : \beta \neq 0$$

Our method

We develop a generalized approach by using latent genetic variables (Z_1, Z_2) to describe the uncertainty of the mode of inheritance.

$$(Z_1, Z_2) = \begin{cases} (0,0) & \textit{when } G = G_0, \\ (0,1) & \textit{when } G = G_1, \\ (1,0) & \textit{when } G = G_2. \end{cases}$$

Then

$$\textit{logit } P(Y = 1|G) = \alpha + \gamma X + \beta Z_1 + \theta\beta Z_2$$

Our method

Note: Under the null hypothesis, the parameter θ is not estimable from the data. Thus, standard statistical tests, such as score test or likelihood-ratio tests are not applicable.

However, for each fixed value of θ, model gives a valid way of testing the null hypothesis. Varying θ, one can get a family of score statistics.

We propose to use the maximum value of such score statistics over a suitable range of θ as the final test statistics.

Our method

- Definations

Assume $\Theta = (\psi, \beta)^T$, $\psi = (\alpha, \gamma)^T$ is nuisance parameters. Let the log-likelihood function is $l(\Theta)$. Denote the MLE as $\hat{\Theta} = (\hat{\psi}, \hat{\beta})^T$ without any restriction. The MLE under H_0 is $\tilde{\Theta} = (\tilde{\psi}, 0)$.

The Score function is

$$U(\Theta) = \begin{bmatrix} U_\beta(\Theta) \\ U_\psi(\Theta) \end{bmatrix} = \begin{bmatrix} \frac{\partial l(\theta)}{\partial \beta} \\ \frac{\partial l(\theta)}{\partial \psi} \end{bmatrix}$$

Our method

The observed Fisher information matrix

$$F_n(\Theta) = -\frac{\partial^2 l(\Theta)}{\partial\Theta\partial\Theta^T} = \begin{bmatrix} f_{\beta\beta}(\Theta) & f_{\beta\psi}(\Theta) \\ f_{\psi\beta}(\Theta) & f_{\psi\psi}(\Theta) \end{bmatrix}$$

The inverse of $F_n(\Theta)$ is

$$F_n^{-1}(\Theta) = \begin{bmatrix} f^{\beta\beta}(\Theta) & f^{\beta\psi}(\Theta) \\ f^{\psi\beta}(\Theta) & f^{\psi\psi}(\Theta) \end{bmatrix}$$

Our method

Construct the score test statistic

$$ST(\theta) = U_\beta^T(\theta) f^{\beta\beta}(\theta) U_\beta(\theta)\Big|_{\tilde{\Theta}}$$

where ST is a function of θ, and then vary the value of θ, the final statistics is

$$Smax = \max_{0 \le \theta \le 1} ST(\theta)$$

Simulation Settings

- We conduct simulation studies to compare our method with standard score test.
- Set $\alpha = -1.2, \gamma = 1$, $\beta = 0.6$ and X are 10000 samples from $N(0,1)$ and 10000 samples from $Binomial(1/2)$, $G(0,1,2)$ are 20000 samples from $Trinomial(g_0, g_1, g_2)$.
- Set the total number of cases and controls are respectively $R = 1000, S = 1000$.
- We consider $\theta = 0, 0.25, 0.5$ as the true genetic models.

Type I Error Rates

Table 2. The empirical type I error rates of our method and the score test

Method	MAF					
	0.05	0.15	0.25	0.35	0.45	0.55
Score	0.043	0.052	0.044	0.049	0.051	0.053
Smax	0.048	0.055	0.051	0.039	0.054	0.060

The estimation of the model

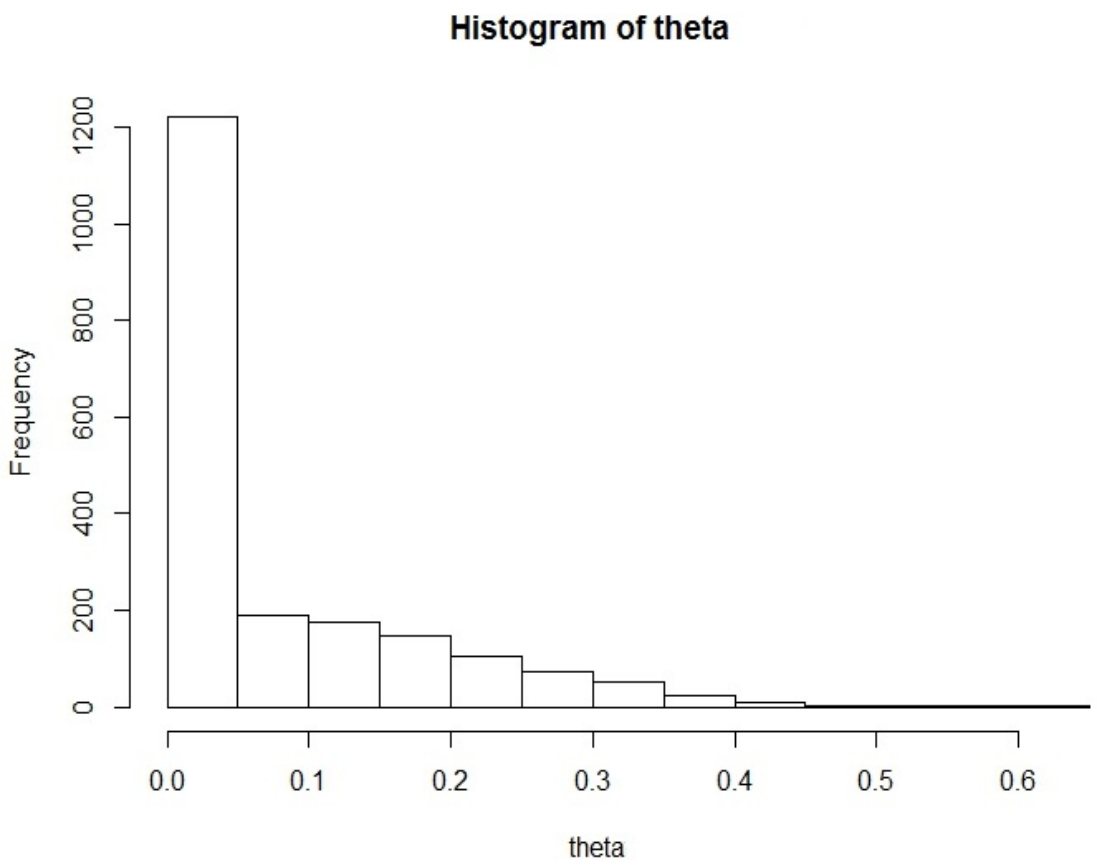

Figure: The estimation of model using our method when the true model is recessive model.

Power Comparison

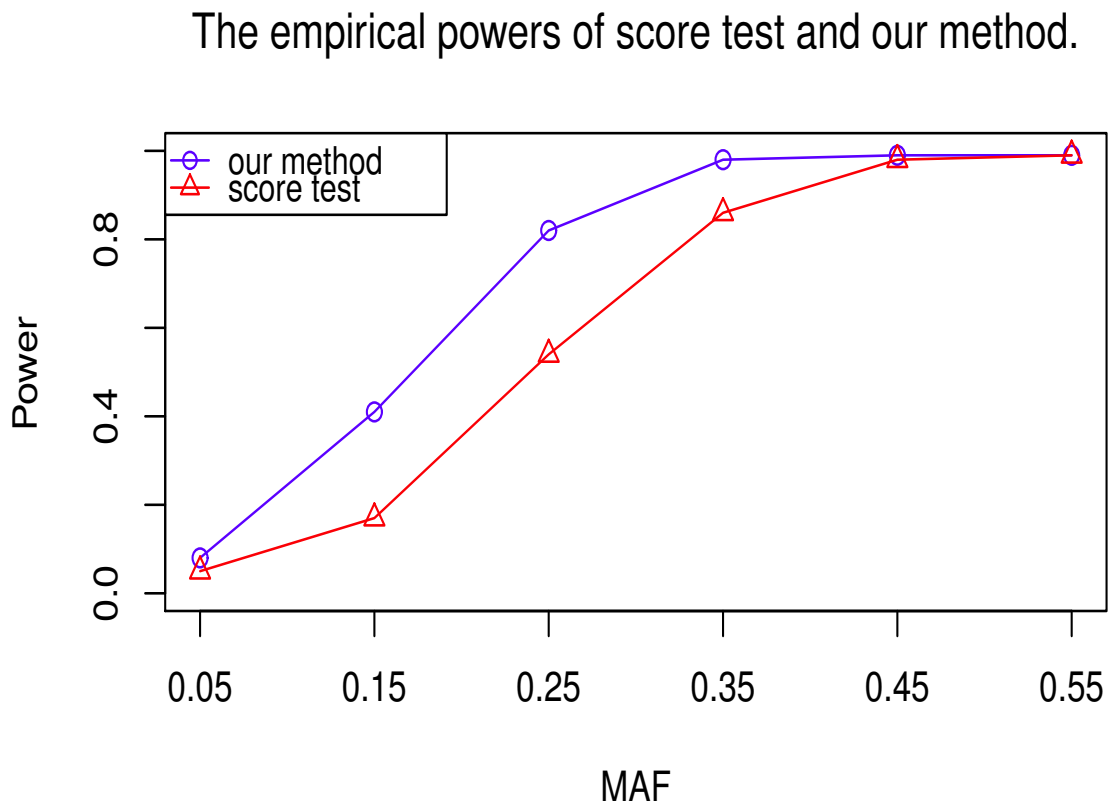

Figure: The empirical powers of our method and the score test that the genetic effects of the SNPs follow a model of $\theta = 0$.

Power Comparison

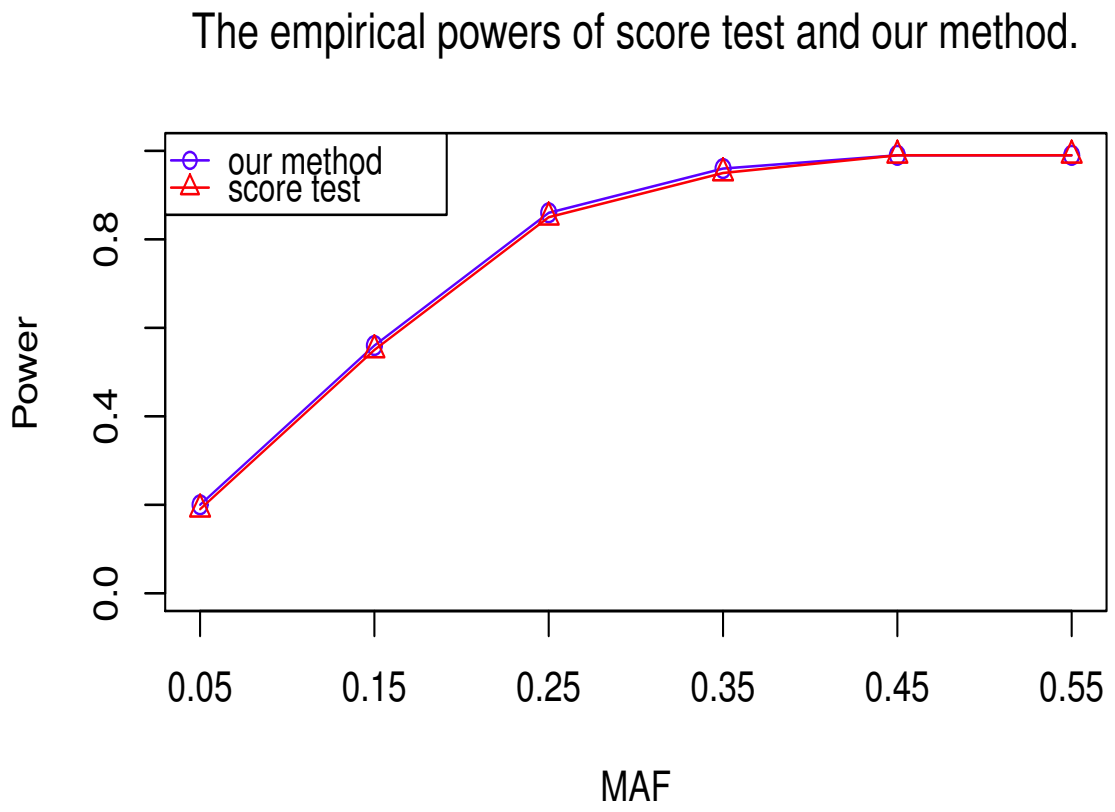

Figure: The empirical powers of our method and the score test that genetic effects of the SNPs follow a model of $\theta = 0.25$.

Power Comparison

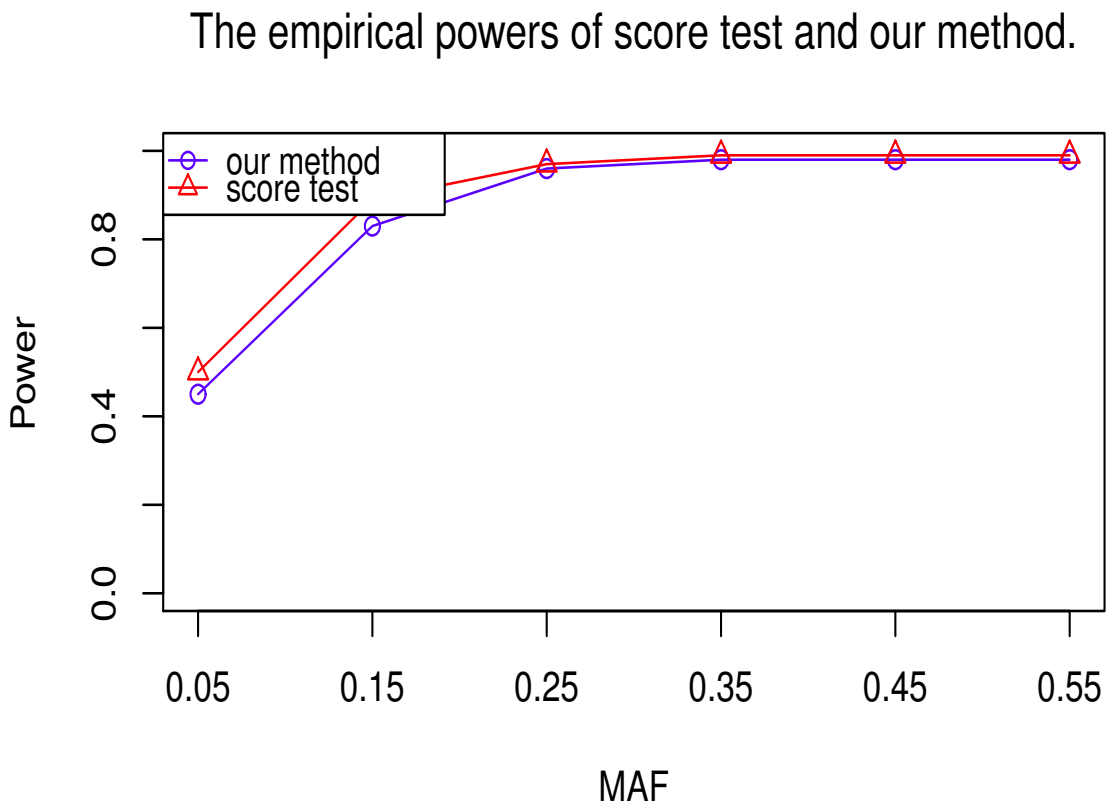

Figure: The empirical powers of our method and the score test that genetic effects of the SNPs follow a model of $\theta = 0.5$.

Future Work

As one or more functional loci exist in the region and interact with the environment factor in a complicated way, it would misrepresent and underestimate the genetic contribution to the joint effect.

We will propose a multilocus test that can simultaneously detect both the main effects and the gene-gene and gene-environment interactions.

THANKS!